Table S1 Search strategy used in PubMed, EMBASE and the Cochrane Library databases

Database	PubMed		EMBASE		The Cochrane Library database	
Search number	Search query	Results	Search query	Results	Search query	Results
#1	methicillin-susceptible *Staphylococcus aureus*	3018	methicillin-susceptible *Staphylococcus aureus* [exp]	6835	methicillin-susceptible *Staphylococcus aureus*	176
#2	methicillin susceptible *Staphylococcus aureus*	5420	methicillin susceptible *Staphylococcus aureus* [exp]	6835	methicillin susceptible *Staphylococcus aureus*	238
#3	methicillin-sensitive *Staphylococcus aureus*	2300	methicillin-sensitive *Staphylococcus aureus* [exp]	6746	methicillin-sensitive *Staphylococcus aureus*	70
#4	methicillin sensitive *Staphylococcus aureus*	3920	methicillin sensitive *Staphylococcus aureus* [exp]	6746	methicillin sensitive *Staphylococcus aureus*	104
#5	MSSA	3257	MSSA	8884	MSSA	113
#6	#1 OR #2 OR #3 OR #4 OR #5	9488	#1 OR #2 OR #3 OR #4 OR #5	9786	#1 OR #2 OR #3 OR #4 OR #5	333
#7	bacteremia [mh]	45061	bacteremia [exp]	56699	bacteremia	2479
#8	bacteraemia	45061	bacteraemia [exp]	49825	bacteraemia	2479
#9	bloodstream infection	27505	bloodstream infection [exp]	15618	bloodstream infection	983
#10	#7 OR #8 OR #9	66994	#7 OR #8 OR #9	70992	#7 OR #8 OR #9	3218
#11	minimum inhibitory concentration	134806	minimum inhibitory concentration [exp]	107738	minimum inhibitory concentration	1152
#12	microbial sensitivity tests [mh]	128403	microbial sensitivity tests [mh]	9307	microbial sensitivity tests	1457
#13	MIC	39349	MIC	128645	MIC	1352
#14	#11 OR #12 OR #13	147130	#11 OR #12 OR #13	137364	#11 OR #12 OR #13	3219
#15	vancomycin [mh]	29888	vancomycin [mh]	95211	vancomycin	2008
#16	#6 AND #10 AND #14 AND #15	205	#6 AND #10 AND #14 AND #15	170	#6 AND #10 AND #14 AND #15	13

Table S2 Raw data of mortality extracted from the included studies

Study	No. MSSA	No. death	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]		Mortality definition
			Events	Total	Events	Total	Unadjusted	Adjusted	
Abelenda 2018	58	15	6	23	10	35	**0.88 (0.27, 2.89)**	NA	30-day
Aguado 2011	99	14	6	23	8	76	**3.00 (0.92, 9.81)**	NA	30-day
Baxi 2016	230	37	11	39	26	191	2.47 (0.99, 8.96)	**1.71 (0.69, 5.35)** [d]	90-day
	230	27	7	39	20	191	1.95 (0.73, 6.75)	1.16 (0.42, 3.75) [d]	30-day
Bouiller 2018	250	64	18	64	46	186	**1.19 (0.63, 2.26)**	NA	30-day
Caston 2014	53	12	8	21	4	32	4.31 (1.10, 16.93)	**9.30 (1.31, 63.2)**	30-day
Cervera 2014	93	40	22	40	18	53	**2.38 (1.02, 5.52)**	NA	One-year
	93	37	21	40	16	53	2.56 (1.09, 6.00)	NA	In-hospital
Fernandez-Hidalgo 2018	173	58	19	60	39	113	0.88 (0.45, 1.72)	**0.87 (0.39, 1.92)**	In-hospital
Han 2012	202	25	NA	NA	NA	NA	1.47 (0.61, 3.54) [b]	**3.90 (1.07, 14.2)**	30-day
Holmes 2011	330	46	NA	NA	NA	NA	2.52 (1.33, 4.78) [b]	**2.44 (1.21, 4.92)**	30-day
Lopez-Cortes 2015	135	37	8	29	29	106	1.01 (0.40, 2.54)	**0.85 (0.32, 3.35)** [c]	30-day
	135	28	3	29	25	106	0.37 (0.10, 1.34)	0.33 (0.09, 1.50) [c]	14-day
San-Juan 2016	83	10	4	43	6	40	**0.58 (0.15, 2.23)**	NA	30-day
Song 2017	334	83	5	27	78	307	0.67 (0.24, 1.82)	NA	30-day (Etest)
	334	83	5	27	78	307	**1.54 (0.51, 4.66)**	NA	30-day (BMD)
Sullivan 2017	252	66	19	84	47	168	**0.75 (0.41, 1.39)**	NA	90-day
	252	59	18	84	41	168	0.84 (0.45, 1.58)	NA	60-day
	252	45	14	84	31	168	0.88 (0.44, 1.77)	NA	30-day
Videma 2014	84	32	25	55	7	29	**2. 62 (0.96, 7.14)**	NA	In-hospital
	84	17	13	55	4	29	1.93 (0.57, 6.59)	NA	30-day

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies unless specified otherwise.

[b] The unadjusted odds ratio and 95% confidence interval were directly extracted from the study.

[c] The adjusted odds ratio and 95% confidence interval were calculated from the study.

Table S3 Raw data of septic thrombophlebitis extracted from the included studies

Study	No. MSSA	No. septic thrombophlebitis	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Aguado 2011	99	13	8	23	5	76	**7.57 (2.17, 26.39)**	NA
Bouiller 2018	250	15	3	64	12	186	**0.71 (0.19, 2.61)**	NA
Caston 2014	53	4	1	21	3	32	**0.48 (0.05, 4.99)**	NA
Cervera 2014	93	19	13	40	6	53	**3.77 (1.28, 11.07)**	NA
Lopez-Cortes 2015	135	2	2	29	0	106	**19.36 (0.90, 415,12)**	NA
San-Juan 2016	49	10	9	43	1	40	**10.32 (1.24, 85.71)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

Table S4 Raw data of persistent bacteraemia extracted from the included studies

Study	No. MSSA	No. persistent bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted [b]	Adjusted
Abelenda 2018	58	16	9	23	7	35	**2.57 (0.79, 8.35)**	NA
Cervera 2014	93	9	4	40	5	53	**1.07 (0.27, 4.26)**	NA
Lopez-Cortes 2015	125	26	8	28	18	97	**1.76 (0.67, 4.62)**	NA
San-Juan 2016	83	6	4	43	2	40	**1.95 (0.34, 11.27)**	NA

[a] Data with bold font were used in the main analysis.

[b] Unadjusted odds ratios and 95% confidence intervals were calculated from the studies

Table S5 Raw data of complicated bacteraemia extracted from the included studies

Study	No. MSSA	No. complicated bacteraemia	High MIC		Low MIC		Odds ratios and 95% confidence intervals [a]	
			Events	Total	Events	Total	Unadjusted	Adjusted
Aguado 2011	99	28	18	23	10	76	23.8 (7.20, 78.4)	**22.9 (6.7, 78.1)**
Bouiller 2018	250	83	10	64	73	186	**0.29 (0.14, 0.60)**	NA
Caston 2014	53	10	2	21	8	32	**0.32 (0.06, 1.66)**	NA
Falcon 2019	113	NA	NA	NA	NA	NA	**1.72 (0.66, 4.48)** [b] (BMD)	NA
	113	NA	NA	NA	NA	NA	1.69 (0.53, 5.38) [b] (Etest)	NA
Lopez-Cortes 2015	135	19	4	29	15	106	**0.97 (0.30, 3.19)**	NA
San-Juan 2016	83	26	18	43	8	40	**2.88 (1.08, 7.70)**	NA
Sullivan 2017	252	50	NA	NA	NA	NA	NA	**2.35 (1.26, 4.37)**

[a] Data with bold font were used in the main analysis. Unadjusted odds ratios and 95% confidence intervals were calculated from the studies unless specified otherwise. Adjusted odds ratios and 95% confidence intervals were directly extracted from the studies.

[b] The unadjusted odds ratio and 95% confidence intervals was directly extracted from the study

Table S6 Assessment of the quality of the included studies

Study	Confounding	Selection of participants	Classification of interventions	Deviations from intended interventions	Missing data	Measurement of outcomes	Selection of the reported result	Overall
Abelenda 2018	Critical ROB	Serious ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Aguado 2011	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Baxi 2016	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
Bouiller 2018	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Caston 2014	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Cervera 2014	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Falcón1 2019	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Fernandez-Hidalgo 2017	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Han 2012	Low ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB
Holmes 2011	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB	Moderate ROB
Lopez-Cortes 2015	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB
San-Juan 2016	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Song 2017	Critical ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Sullivan 2017	Critical ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Critical ROB
Viedma 2014	Moderate ROB	Moderate ROB	Low ROB	Low ROB	Low ROB	Low ROB	Low ROB	Moderate ROB

Abbreviations: ROB, risk of bias.

Table S7 Leave-one-out sensitivity analysis of mortality

Study omitted	OR (95% CI)	P Value	I^2 %
Abelenda 2018	1.47 (1.14-1.89)	0.003	43.1
Aguado 2011	1.39 (1.08-1.79)	0.01	40.7
Baxi 2016	1.42 (1.10-1.83)	0.007	44.6
Bouiller 2018	1.48 (1.14-1.94)	0.004	43.9
Caston 2014	1.39 (1.09-1.79)	0.009	33.9
Cervera 2014	1.37 (1.06-1.77)	0.016	40.8
Fernandez-Hidalgo 2018	1.52 (1.17-1.96)	0.002	40.3
Han 2012	1.38 (1.08-1.78)	0.011	38.2
Holmes 2011	1.33 (1.03-1.74)	0.032	37.8
Lopez-Cortes 2015	1.47 (1.14-1.89)	0.003	42.8
San-Juan 2016	1.48 (1.15-1.90)	0.002	40.0
Song 2017	1.43 (1.11-1.84)	0.005	44.9
Sullivan 2017	1.63 (1.25-2.13)	< 0.001	27.6
Viedma 2014	1.38 (1.07-1.78)	0.013	40.9

Table S8 Definitions of the secondary outcomes

Study	Outcomes	Definitions of the secondary outcomes
Abelenda 2018 [13]	PB	PB was defined as positive blood cultures after more than 48 hours of appropriate antibiotic therapy.
Aguado. 2011 [8]	ST, CB	ST was defined by persistent bacteraemia at least 72 hours after initiation of active antimicrobial drugs and documented thrombi. CB was defined by one of the following events occurring after the first episode of bacteraemia: 1) development of endocarditis, septic thrombophlebitis, arthritis, spondylitis, as well as end-organ hematogenous spread of infection to other locations; 2) infection involving vascular or osteoarticular prostheses (excluding intravascular catheter) not removed within 4 days.
Bouiller 2018 [15]	ST, CB	The definition of ST was not mentioned. The definition of CB was the same as Aguado et al [8].
Caston 2014 [16]	ST, CB	The definitions of ST and CB were not mentioned.
Cervera 2014 [17]	ST, PB	The definition of ST was not mentioned. PB was defined as positive blood cultures ≥ 7 days after antimicrobial treatment.
Falcon 2019 [18]	CB	CB was defined by the following events: 1) endocarditis; 2) septic thrombophlebitis; 3) metastatic infection; 4) persistent bacteraemia for > 72 h after initiating active antimicrobial drug therapy, and 5) recurrent bacteraemia.
Lopez-Cortes 2015 [21]	ST, PB, CB	The definition of ST and CB were the same as Aguado et al [8]. PB was defined as the isolation of *S. aureus* in blood cultures obtained from peripheral veins for ≥ 3 days.
San-Juan 2016 [22]	ST, PB, CB	ST was defined by persistent bacteraemia for > 72 hours after the initiation of active therapy associated with the documentation of a thrombus at the site of catheter insertion. PB was defined as positive blood cultures after > 72 hours of appropriate antibiotic therapy. CB was defined by one of five events occurring after the incident episode of MSSA CRBSI: 1) development of endocarditis; 2) septic thrombophlebitis; 3) septic metastatic infection, including arthritis, spondylitis, infections involving vascular or osteoarticular prostheses (excluding intravascular catheter) or end-organ hematogenous seeding to other locations remote from the primary focus; 4) persistent bacteraemia for > 72 hours; or 5) persistent fever for > 4 days after extraction of the first blood culture yielding MSSA, provided that alternative causes had been reasonably excluded.
Sullivan 2017 [24]	CB	Complicated bacteraemia was defined by metastatic or embolic seeding, endovascular infection, or a duration > 3 days.

Abbreviations: CB, complicated bacteraemia; CRBSI, catheter-related bloodstream infections; MSSA, methicillin-susceptible *Staphylococcus aureus*; PB, persistent bacteraemia; SB, septic thrombophlebitis.

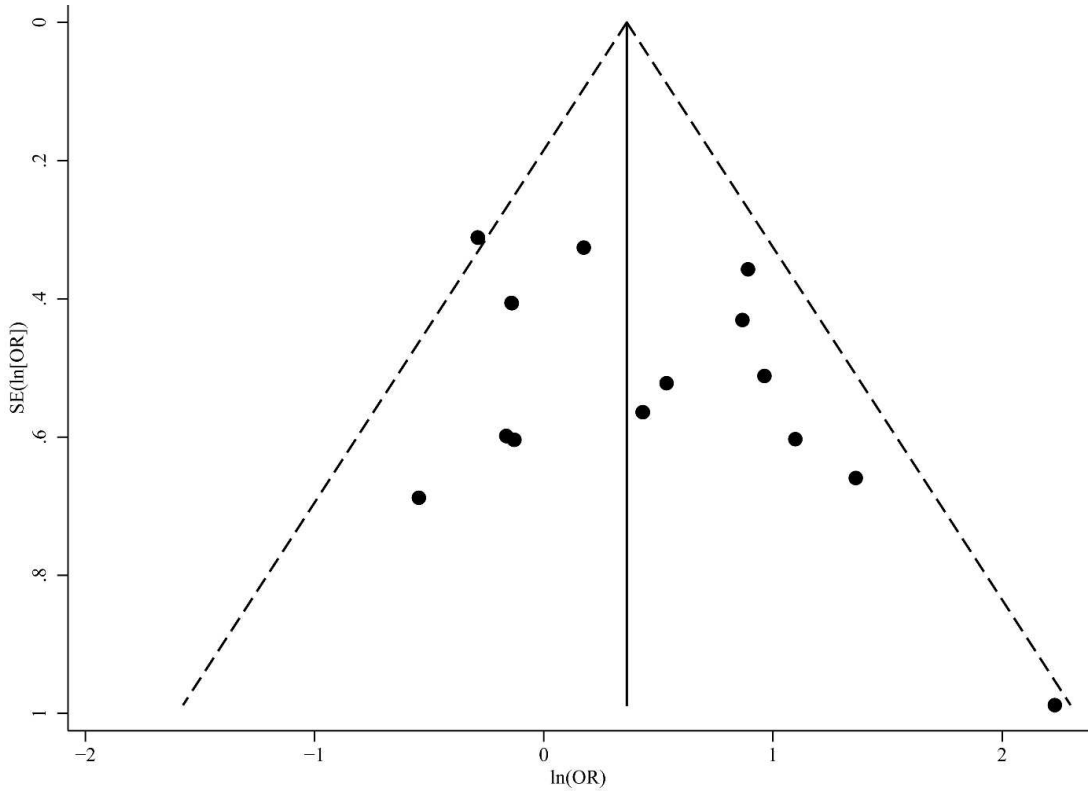


Figure S1 Funnel plot of mortality.